Blessed G/Michael Catholic Secondary School

Semester 1 Biology Mid1 Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** In a monohybrid cross between two heterozygous pea plants (Tt x Tt), what is the expected phenotypic ratio of tall to dwarf offspring?

A. 1 : 1 **B.** 3 : 1 **C.** 9 : 3 : 3 : 1 **D.** 1 : 2 : 1

____ **2.** Which enzyme is responsible for synthesizing new complementary DNA strands during semi-conservative replication?

A. RNA Polymerase **B.** DNA Helicase **C.** DNA Polymerase **D.** DNA Ligase

____ **3.** The synthesis of mRNA from a DNA template strand in the cell nucleus is called:

A. Translation **B.** Transcription **C.** Transduction **D.** Transformation

____ **4.** Human ABO blood groups demonstrate which non-Mendelian genetic inheritance pattern?

A. Incomplete dominance **B.** Codominance and multiple alleles

C. Polygenic inheritance **D.** Sex linkage

____ **5.** Red-green color blindness in humans is inherited as an:

A. Autosomal dominant trait **B.** Autosomal recessive trait

C. X-linked recessive trait **D.** Y-linked trait

____ **6.** Which nitrogenous base is found in RNA molecules but completely absent in DNA?

A. Thymine **B.** Uracil **C.** Adenine **D.** Cytosine

____ **7.** Mendel's Law of Independent Assortment states that:

A. Alleles of different unlinked genes segregate independently during gamete formation
B. Dominant alleles mask recessive alleles
C. Homologous chromosomes never separate
D. Genes on the same chromosome always assort together

____ **8.** In a dihybrid cross between two heterozygous individuals (AaBb x AaBb), what is the classic Mendelian phenotypic ratio?

A. 9 : 3 : 3 : 1 **B.** 3 : 1 **C.** 1 : 2 : 1 **D.** 15 : 1

____ **9.** What is the phenotypic outcome of incomplete dominance in snapdragon flower color when crossing red (RR) with white (rr)?

A. 100% Pink flowers (Rr) **B.** 100% Red flowers

C. 50% Red and 50% White **D.** 100% White flowers

____ **10.** Which genetic disorder is caused by trisomy of chromosome 21?

A. Down syndrome **B.** Turner syndrome

C. Klinefelter syndrome **D.** Edwards syndrome

____ **11.** Which DNA sequence marks the binding site for RNA polymerase to initiate gene transcription?

A. Promoter region **B.** Terminator region

C. Operator **D.** Intron

____ **12.** During mRNA processing in eukaryotes, which non-coding sequences are excised from pre-mRNA transcripts?

A. Introns **B.** Exons **C.** Codons **D.** Promoters

____ **13.** The universal genetic start codon that codes for methionine in mRNA translation is:

A. AUG **B.** UAA **C.** UAG **D.** UGA

____ **14.** Which of the following is a stop (termination) codon that halts peptide synthesis?

A. UAA, UAG, and UGA **B.** AUG, GUG, and UUG

C. AAA, UUU, and CCC **D.** CAG and GAC

____ **15.** Hemophilia and Duchenne muscular dystrophy are classic examples of:

A. X-linked recessive disorders **B.** Autosomal dominant disorders

C. Y-linked holandric disorders **D.** Mitochondrial disorders

____ **16.** The genetic code is described as "degenerate" (redundant) because:

A. Multiple different codons can code for the exact same amino acid
B. One codon can specify multiple different amino acids
C. It deteriorates over time
D. It contains no punctuation

____ **17.** In DNA replication, Okazaki fragments are synthesized discontinuously on the:

A. Lagging strand (in 5' to 3' direction) **B.** Leading strand

C. Template strand **D.** RNA primer strand

____ **18.** Which enzyme seals nicks in the phosphodiester backbone by joining Okazaki fragments during DNA replication?

A. DNA Ligase **B.** DNA Helicase **C.** Topoisomerase **D.** Primase

____ **19.** A point mutation that replaces a single nucleotide resulting in a premature STOP codon is called a:

A. Nonsense mutation **B.** Missense mutation

C. Silent mutation **D.** Frameshift mutation

____ **20.** What is a Test Cross used for in genetics?

A. Crossing an organism of dominant phenotype with a homozygous recessive individual to determine its genotype
B. Crossing two homozygous dominant individuals
C. Testing for sterility
D. Crossing two different species

____ **21.** The Hardy-Weinberg equilibrium equation for allele frequencies in an idealized non-evolving population is:

A. $p^2 + 2pq + q^2 = 1$ **B.** $p + q = 2$

C. $p^2 * q^2 = 1$ **D.** $p - q = 0$

____ **22.** Which type of RNA forms the structural core and catalytic peptidyl transferase center of ribosomes?

A. rRNA (ribosomal RNA) **B.** mRNA (messenger RNA)

C. tRNA (transfer RNA) **D.** siRNA

____ **23.** Recombinant DNA technology commonly uses which bacterial enzymes to cut DNA at specific palindromic recognition sequences?

A. Restriction endonucleases **B.** DNA polymerases

C. Reverse transcriptases **D.** Amylases

____ **24.** Polymerase Chain Reaction (PCR) is a molecular biology laboratory technique used to:

A. Amplify millions of copies of a specific targeted DNA sequence in vitro
B. Synthesize proteins directly
C. Sequester carbohydrates
D. Stain cellular cytoplasm

____ **25.** Which of the following karyotypes represents Turner syndrome in females?

A. 45, X0 (monosomy X) **B.** 47, XXY

C. 47, XYY **D.** 46, XY